

DPT Exercises

Exercise sheet 11

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Exercise 1 (hypothesis test for a mean). A factory produces light bulbs, and the historical average lifetime is $\mu_0 = 1200$ hours with a known standard deviation of $\sigma = 150$ hours. A new manufacturing process is introduced, and the manager wants to test if the average lifetime has changed. A sample of $n = 36$ light bulbs produced by the new process has a sample mean lifetime of $\bar{x}_n = 1250$ hours. Assume the lifetimes are normally distributed.

- (i) State the null hypothesis H_0 and the alternative hypothesis H_1 for this test.
- (ii) Calculate the value of the appropriate test statistic.
- (iii) Would you reject the null at a significance level of $\alpha = 0.05$? What about at $\alpha = 0.001$? What does that mean for the confidence you should have in the rejection?

Solution:

- (i) The null hypothesis is that the average lifetime has not changed from the historical average: $H_0 : \mu = \mu_0 = 1200$. The alternative hypothesis is that the average lifetime has changed (two-sided test): $H_1 : \mu \neq 1200$.
- (ii) Since the population standard deviation σ is known and lifetimes are assumed normal, we use a z-test statistic

$$Z = \frac{\bar{X}_n - \mu_0}{\sigma/\sqrt{n}}.$$

The observed value is:

$$z_{\text{obs}} = \frac{1250 - 1200}{150/\sqrt{36}} = \frac{50}{150/6} = \frac{50}{25} = 2.$$

- (iii) We reject when $|z_{\text{obs}}| \geq z_{\alpha/2}$ where $z_{\alpha/2}$ is the $1 - \alpha/2$ -quantile of a standard normal, which is approximately 1.96 for $\alpha = 0.05$. Since $2 > 1.96$ we reject the null at the 5% significance level. However, since $z_{\alpha/2} \approx 3.96 > 2$ for $\alpha = 0.001$, we do not reject the hypothesis at significance level 1%. This does not mean that we should be 95% certain that there was an improvement, but cannot be 99% certain. The significance level is $\alpha = \mathbb{E}_{\mu_0}[\varphi]$, i.e., an expectation (probability) over “repeating the procedure on new datasets.” Given just a single dataset, it does directly give us a meaningful “confidence.”

Exercise 2 (Neyman-Pearson test for exponential distribution). Suppose we observe a single sample X from an exponential distribution $\text{Exp}(\lambda)$. We want to test the simple null hypothesis $H_0 : \{\lambda_0\}$ against the simple alternative hypothesis $H_1 : \{\lambda_1\}$, where $\lambda_1 > \lambda_0 > 0$.

- (i) Write down the likelihood ratio $\Lambda(x) = \frac{L(\lambda_1|x)}{L(\lambda_0|x)}$.
- (ii) According to the Neyman-Pearson lemma, the most powerful test of size α rejects H_0 if $\Lambda(x) > k'$ for some constant k' . Show that this rejection region is equivalent to $x < c$ or $x > c$ for some constant c . Determine which form is correct for $\lambda_1 > \lambda_0$.
- (iii) For a significance level α , determine the explicit form of the critical value c for the rejection region.
- (iv) What is the power of this test?

Solution:

- (i) The pdf of $\text{Exp}(\lambda)$ is $p(x | \lambda) = \lambda e^{-\lambda x}$ for $x \geq 0$. For a single observation x , the likelihood is $L(\lambda | x) = \lambda e^{-\lambda x}$. The likelihood ratio is:

$$\Lambda(x) = \frac{L(\lambda_1 | x)}{L(\lambda_0 | x)} = \frac{\lambda_1 e^{-\lambda_1 x}}{\lambda_0 e^{-\lambda_0 x}} = \frac{\lambda_1}{\lambda_0} e^{-(\lambda_1 - \lambda_0)x}.$$

- (ii) The rejection region is $\Lambda(x) > k'$ for some $k' > 0$.

$$\frac{\lambda_1}{\lambda_0} e^{-(\lambda_1 - \lambda_0)x} > k' \implies e^{-(\lambda_1 - \lambda_0)x} > k' \frac{\lambda_0}{\lambda_1}.$$

Let $k'' = k' \frac{\lambda_0}{\lambda_1}$. Since $\lambda_0, \lambda_1, k' > 0$, we have $k'' > 0$. Taking $\ln$ on both sides (which is well-defined and order-preserving as $\ln$ is strictly increasing):

$$-(\lambda_1 - \lambda_0)x > \ln(k'').$$

Since $\lambda_1 > \lambda_0$, the term $\lambda_1 - \lambda_0 > 0$. Dividing by $-(\lambda_1 - \lambda_0)$ (which is negative) flips the inequality sign:

$$x < -\frac{\ln(k'')}{\lambda_1 - \lambda_0}.$$

Let $c = -\frac{\ln(k'')}{\lambda_1 - \lambda_0}$. The rejection region is of the form $x < c$. This makes intuitive sense: if we observe a small value of x , it is more likely to come from an exponential distribution with a larger rate λ_1 (since higher rate means faster decay, smaller expected value $1/\lambda$). The value c must be positive for the test to be non-trivial (otherwise $P_{\lambda_0}(X < c) = 0$ if $c \leq 0$). This depends on k'' . If $\ln(k'') > 0$ (i.e., $k'' > 1$), then $c < 0$. If $\ln(k'') < 0$ (i.e., $0 < k'' < 1$), then $c > 0$.

- (iii) For the significance level we compute

$$\alpha = P_{\lambda_0}(X < c) = \int_0^c \lambda_0 e^{-\lambda_0 x} dx = [-e^{-\lambda_0 x}]_0^c = 1 - e^{-\lambda_0 c}.$$

Solving for c yields

$$e^{-\lambda_0 c} = 1 - \alpha \implies -\lambda_0 c = \ln(1 - \alpha) \implies c = -\frac{\ln(1 - \alpha)}{\lambda_0}.$$

Since $0 < \alpha < 1$, $\ln(1 - \alpha) < 0$, so $c > 0$. The test rejects H_0 if $x < -\frac{\ln(1 - \alpha)}{\lambda_0}$.

- (iv) For the power of the test at α , we compute

$$P_{\lambda_1}(X < c) = \int_0^c \lambda_1 e^{-\lambda_1 x} dx = 1 - e^{-\lambda_1 c}.$$

Substituting $c = -\frac{\ln(1 - \alpha)}{\lambda_0}$, we end up with

$$G_{\varphi}(\lambda_1) = 1 - e^{-\lambda_1 \left(-\frac{\ln(1 - \alpha)}{\lambda_0}\right)} = 1 - e^{\frac{\lambda_1}{\lambda_0} \ln(1 - \alpha)} = 1 - (e^{\ln(1 - \alpha)})^{\lambda_1 / \lambda_0} = 1 - (1 - \alpha)^{\lambda_1 / \lambda_0}.$$

Exercise 3 (one-sample t-test for a mean). A brewery claims that its bottles contain exactly 330 ml of beer. A skeptical student measures the contents of $n = 10$ randomly selected bottles and finds a sample mean of $\bar{x}_{10} = 328.5$ ml and a sample variance of $s_{10}^2 = 4$ ml². Assuming the amount of beer in a bottle is normally distributed.

- (i) State the null and alternative hypotheses for testing if the true mean amount of beer is different from 330 ml.

- (ii) What is the appropriate test for this situation? State the test statistic and its distribution under the null hypothesis.
- (iii) Perform the test at a significance level of $\alpha = 0.05$. Should the student reject the brewery's claim? [Hint: You may use that the 0.975-quantile of a t -distribution with 9 degrees of freedom is $t_{9,0.025} \approx 2.262$.]

Solution:

- (i) The null and alternative hypotheses are:

- Null hypothesis $H_0 : \mu = 330$. The true mean amount of beer is 330 ml.
- Alternative hypothesis $H_1 : \mu \neq 330$. The true mean amount is not 330 ml.

This is a two-sided test.

- (ii) Since the population variance σ^2 is unknown and we are testing the mean of a normally distributed population, the appropriate test is a one-sample t -test. The test statistic is

$$T = \frac{\bar{X}_n - \mu_0}{S_n / \sqrt{n}},$$

where $\mu_0 = 330$ is the mean under the null hypothesis, and $S_n = \sqrt{S_n^2}$ is the sample standard deviation. Under H_0 , this statistic follows a Student's t -distribution with $n - 1 = 9$ degrees of freedom.

- (iii) We calculate the observed value of the test statistic t_{obs} . The sample standard deviation is $s_{10} = \sqrt{4} = 2$.

$$t_{\text{obs}} = \frac{328.5 - 330}{2/\sqrt{10}} = \frac{-1.5}{2/3.162} \approx \frac{-1.5}{0.632} \approx -2.373.$$

For a two-sided test at level $\alpha = 0.05$, the rejection region is $|t_{\text{obs}}| > t_{n-1, \alpha/2} = t_{9, 0.025}$. The critical value is given as $t_{9, 0.025} \approx 2.262$.

Since $|t_{\text{obs}}| = |-2.373| = 2.373$, and $2.373 > 2.262$, the observed test statistic falls into the rejection region. Therefore, the student should reject the null hypothesis. There is significant evidence at the 5% level to suggest that the mean volume of the beer bottles is not 330 ml.

Exercise 4 (type II error and power of a z -test). Suppose you are testing the mean of a normal population $\mathcal{N}(\mu, \sigma^2)$ with known variance $\sigma^2 = 9$. You collect $n = 25$ samples. You want to test the null hypothesis $H_0 : \mu \leq 10$ against the alternative $H_1 : \mu > 10$ at a significance level of $\alpha = 0.01$.

- (i) Determine the rejection region for this test. You may use that the 0.99-quantile of a standard normal distribution is $z_{0.01} \approx 2.33$.
- (ii) Calculate the probability of a type II error, β , if the true mean is actually $\mu_a = 11.5$.
- (iii) What is the power of the test against the alternative $\mu_a = 11.5$?

Solution:

- (i) We use a one-sided z -test. The test statistic under the null boundary $\mu_0 = 10$ is

$$Z = \frac{\bar{X}_n - \mu_0}{\sigma / \sqrt{n}} = \frac{\bar{X}_{25} - 10}{3 / \sqrt{25}} = \frac{\bar{X}_{25} - 10}{0.6}.$$

Under H_0 , $Z \sim \mathcal{N}(0, 1)$. For a one-sided test at level $\alpha = 0.01$, we reject H_0 if $z_{\text{obs}} > z_{0.01} \approx 2.33$. The rejection region in terms of the sample mean $\bar{X}_{25}$ is

$$\frac{\bar{X}_{25} - 10}{0.6} > 2.33 \implies \bar{X}_{25} > 10 + 2.33 \cdot 0.6 = 10 + 1.398 = 11.398.$$

So we reject H_0 if the sample mean is greater than 11.398.

- (ii) A type II error occurs if we fail to reject H_0 when H_1 is true. Here, we calculate this probability for the specific case where the true mean is $\mu_a = 11.5$.

$$\beta = "P(\text{fail to reject } H_0 \mid \mu = \mu_a)" = P_{\mu_a}(\bar{X}_{25} \leq 11.398) .$$

When $\mu = 11.5$, the sample mean $\bar{X}_{25}$ is distributed as $\mathcal{N}(11.5, \sigma^2/n) = \mathcal{N}(11.5, 9/25) = \mathcal{N}(11.5, 0.36)$. The standard deviation is $\sqrt{0.36} = 0.6$. We standardize the variable to compute the probability:

$$\begin{aligned}\beta &= P\left(\frac{\bar{X}_{25} - 11.5}{0.6} \leq \frac{11.398 - 11.5}{0.6}\right) \\ &= P\left(Z' \leq \frac{-0.102}{0.6}\right) \\ &= P(Z' \leq -0.17) ,\end{aligned}$$

where $Z' \sim \mathcal{N}(0, 1)$. Using the cdf Φ of the standard normal distribution, $\beta = \Phi(-0.17)$. From a standard normal table or software, $\Phi(-0.17) \approx 0.4325$. The probability of a type II error is approximately 43.25%.

- (iii) The power of the test is the probability of correctly rejecting H_0 when H_1 is true:

$$1 - \beta = 1 - 0.4325 = 0.5675 .$$

The power of the test to detect a true mean of $\mu_a = 11.5$ is approximately 56.75%.